

Chapter 1

Introduction

This chapter, introduction, opens a new part of the work and explains how it is organised.

The bus depot sensor was visited on a foggy morning in week 2 of the campaign, and the filters were changed before the foggy readings of that visit were written down.

Notes kept at the bus depot on that foggy day belong to section 1 of the logbook.

The old market sensor was visited on a humid morning in week 3 of the campaign, and the filters were changed before the humid readings of that visit were written down.

Notes kept at the old market on that humid day belong to section 1 of the logbook.

The school yard sensor was visited on a dry morning in week 4 of the campaign, and the filters were changed before the dry readings of that visit were written down.

Notes kept at the school yard on that dry day belong to section 1 of the logbook.

The ring road sensor was visited on a rainy morning in week 5 of the campaign, and the filters were changed before the rainy readings of that visit were written down.

Notes kept at the ring road on that rainy day belong to section 1 of the logbook.

The rail yard sensor was visited on a cold morning in week 6 of the campaign, and the filters were changed before the cold readings of that visit were written down.

Notes kept at the rail yard on that cold day belong to section 2 of the logbook.

The east park sensor was visited on a mild morning in week 7 of the campaign, and the filters were changed before the mild readings of that visit were written down.

Notes kept at the east park on that mild day belong to section 2 of the logbook.

The harbour wall sensor was visited on a dusty morning in week 8 of the campaign, and the filters were changed before the dusty readings of that visit were written down.

Notes kept at the harbour wall on that dusty day belong to section 2 of the logbook.

The town hall sensor was visited on a calm morning in week 9 of the campaign, and the filters were changed before the calm readings of that visit were written down.

Notes kept at the town hall on that calm day belong to section 2 of the logbook.

The north gate sensor was visited on a windy morning in week 10 of the campaign, and the filters were changed before the windy readings of that visit were written down.

Notes kept at the north gate on that windy day belong to section 2 of the logbook.

The river bridge sensor was visited on a foggy morning in week 11 of the campaign, and the filters were changed before the foggy readings of that visit were written down.

Notes kept at the river bridge on that foggy day belong to section 2 of the logbook.

Chapter 2

Background

This chapter, background, opens a new part of the work and explains how it is organised.

The old market sensor was visited on a dry morning in week 13 of the campaign, and the filters were changed before the dry readings of that visit were written down.

Notes kept at the old market on that dry day belong to section 3 of the logbook.

The school yard sensor was visited on a rainy morning in week 14 of the campaign, and the filters were changed before the rainy readings of that visit were written down.

Notes kept at the school yard on that rainy day belong to section 3 of the logbook.

The ring road sensor was visited on a cold morning in week 15 of the campaign, and the filters were changed before the cold readings of that visit were written down.

Notes kept at the ring road on that cold day belong to section 3 of the logbook.

The rail yard sensor was visited on a mild morning in week 16 of the campaign, and the filters were changed before the mild readings of that visit were written down.

Notes kept at the rail yard on that mild day belong to section 3 of the logbook.

The east park sensor was visited on a dusty morning in week 17 of the campaign, and the filters were changed before the dusty readings of that visit were written down.

Notes kept at the east park on that dusty day belong to section 3 of the logbook.

The harbour wall sensor was visited on a calm morning in week 18 of the campaign, and the filters were changed before the calm readings of that visit were written down.

Notes kept at the harbour wall on that calm day belong to section 4 of the logbook.

The town hall sensor was visited on a windy morning in week 19 of the campaign, and the filters were changed before the windy readings of that visit were written down.

Notes kept at the town hall on that windy day belong to section 4 of the logbook.

The north gate sensor was visited on a foggy morning in week 20 of the campaign, and the filters were changed before the foggy readings of that visit were written down.

Notes kept at the north gate on that foggy day belong to section 4 of the logbook.

The river bridge sensor was visited on a humid morning in week 21 of the campaign, and the filters were changed before the humid readings of that visit were written down.

Notes kept at the river bridge on that humid day belong to section 4 of the logbook.

The bus depot sensor was visited on a dry morning in week 22 of the campaign, and the filters were changed before the dry readings of that visit were written down.

Notes kept at the bus depot on that dry day belong to section 4 of the logbook.

The old market sensor was visited on a rainy morning in week 23 of the campaign, and the filters were changed before the rainy readings of that visit were written down.

Notes kept at the old market on that rainy day belong to section 4 of the logbook.

The school yard sensor was visited on a cold morning in week 24 of the campaign, and the filters were changed before the cold readings of that visit were written down.

Notes kept at the school yard on that cold day belong to section 5 of the logbook.

The ring road sensor was visited on a mild morning in week 25 of the campaign, and the filters were changed before the mild readings of that visit were written down.

Notes kept at the ring road on that mild day belong to section 5 of the logbook.

The rail yard sensor was visited on a dusty morning in week 26 of the campaign, and the filters were changed before the dusty readings of that visit were written down.

Notes kept at the rail yard on that dusty day belong to section 5 of the logbook.

The east park sensor was visited on a calm morning in week 27 of the campaign, and the filters were changed before the calm readings of that visit were written down.

Notes kept at the east park on that calm day belong to section 5 of the logbook.

The harbour wall sensor was visited on a windy morning in week 28 of the campaign, and the filters were changed before the windy readings of that visit were written down.

Notes kept at the harbour wall on that windy day belong to section 5 of the logbook.

The town hall sensor was visited on a foggy morning in week 29 of the campaign, and the filters were changed before the foggy readings of that visit were written down.

Notes kept at the town hall on that foggy day belong to section 5 of the logbook.

Chapter 3

System Design

This chapter, system design, opens a new part of the work and explains how it is organised.

The river bridge sensor was visited on a dry morning in week 31 of the campaign, and the filters were changed before the dry readings of that visit were written down.

Notes kept at the river bridge on that dry day belong to section 6 of the logbook.

The bus depot sensor was visited on a rainy morning in week 32 of the campaign, and the filters were changed before the rainy readings of that visit were written down.

Notes kept at the bus depot on that rainy day belong to section 6 of the logbook.

The old market sensor was visited on a cold morning in week 33 of the campaign, and the filters were changed before the cold readings of that visit were written down.

Notes kept at the old market on that cold day belong to section 6 of the logbook.

The school yard sensor was visited on a mild morning in week 34 of the campaign, and the filters were changed before the mild readings of that visit were written down.

Notes kept at the school yard on that mild day belong to section 6 of the logbook.

The ring road sensor was visited on a dusty morning in week 35 of the campaign, and the filters were changed before the dusty readings of that visit were written down.

Notes kept at the ring road on that dusty day belong to section 6 of the logbook.

The rail yard sensor was visited on a calm morning in week 36 of the campaign, and the filters were changed before the calm readings of that visit were written down.

Notes kept at the rail yard on that calm day belong to section 7 of the logbook.

The east park sensor was visited on a windy morning in week 37 of the campaign, and the filters were changed before the windy readings of that visit were written down.

Notes kept at the east park on that windy day belong to section 7 of the logbook.

The harbour wall sensor was visited on a foggy morning in week 38 of the campaign, and the filters were changed before the foggy readings of that visit were written down.

Notes kept at the harbour wall on that foggy day belong to section 7 of the logbook.

The town hall sensor was visited on a humid morning in week 39 of the campaign, and the filters were changed before the humid readings of that visit were written down.

Notes kept at the town hall on that humid day belong to section 7 of the logbook.

The north gate sensor was visited on a dry morning in week 40 of the campaign, and the filters were changed before the dry readings of that visit were written down.

Notes kept at the north gate on that dry day belong to section 7 of the logbook.

The river bridge sensor was visited on a rainy morning in week 41 of the campaign, and the filters were changed before the rainy readings of that visit were written down.

Notes kept at the river bridge on that rainy day belong to section 7 of the logbook.

The bus depot sensor was visited on a cold morning in week 42 of the campaign, and the filters were changed before the cold readings of that visit were written down.

Notes kept at the bus depot on that cold day belong to section 8 of the logbook.

The old market sensor was visited on a mild morning in week 43 of the campaign, and the filters were changed before the mild readings of that visit were written down.

Notes kept at the old market on that mild day belong to section 8 of the logbook.

The school yard sensor was visited on a dusty morning in week 44 of the campaign, and the filters were changed before the dusty readings of that visit were written down.

Notes kept at the school yard on that dusty day belong to section 8 of the logbook.

The ring road sensor was visited on a calm morning in week 45 of the campaign, and the filters were changed before the calm readings of that visit were written down.

Notes kept at the ring road on that calm day belong to section 8 of the logbook.

The rail yard sensor was visited on a windy morning in week 46 of the campaign, and the filters were changed before the windy readings of that visit were written down.

Notes kept at the rail yard on that windy day belong to section 8 of the logbook.

The east park sensor was visited on a foggy morning in week 47 of the campaign, and the filters were changed before the foggy readings of that visit were written down.

Notes kept at the east park on that foggy day belong to section 8 of the logbook.

Chapter 4

Evaluation

This chapter, evaluation, opens a new part of the work and explains how it is organised.

The town hall sensor was visited on a dry morning in week 49 of the campaign, and the filters were changed before the dry readings of that visit were written down.

Notes kept at the town hall on that dry day belong to section 9 of the logbook.

The north gate sensor was visited on a rainy morning in week 50 of the campaign, and the filters were changed before the rainy readings of that visit were written down.

Notes kept at the north gate on that rainy day belong to section 9 of the logbook.

The river bridge sensor was visited on a cold morning in week 51 of the campaign, and the filters were changed before the cold readings of that visit were written down.

Notes kept at the river bridge on that cold day belong to section 9 of the logbook.

The bus depot sensor was visited on a mild morning in week 52 of the campaign, and the filters were changed before the mild readings of that visit were written down.

Notes kept at the bus depot on that mild day belong to section 9 of the logbook.

The old market sensor was visited on a dusty morning in week 53 of the campaign, and the filters were changed before the dusty readings of that visit were written down.

Notes kept at the old market on that dusty day belong to section 9 of the logbook.

The school yard sensor was visited on a calm morning in week 54 of the campaign, and the filters were changed before the calm readings of that visit were written down.

Notes kept at the school yard on that calm day belong to section 10 of the logbook.

The ring road sensor was visited on a windy morning in week 55 of the campaign, and the filters were changed before the windy readings of that visit were written down.

Notes kept at the ring road on that windy day belong to section 10 of the logbook.

The rail yard sensor was visited on a foggy morning in week 56 of the campaign, and the filters were changed before the foggy readings of that visit were written down.

Notes kept at the rail yard on that foggy day belong to section 10 of the logbook.

The east park sensor was visited on a humid morning in week 57 of the campaign, and the filters were changed before the humid readings of that visit were written down.

Notes kept at the east park on that humid day belong to section 10 of the logbook.

The harbour wall sensor was visited on a dry morning in week 58 of the campaign, and the filters were changed before the dry readings of that visit were written down.

Notes kept at the harbour wall on that dry day belong to section 10 of the logbook.

The town hall sensor was visited on a rainy morning in week 59 of the campaign, and the filters were changed before the rainy readings of that visit were written down.

Notes kept at the town hall on that rainy day belong to section 10 of the logbook.

The north gate sensor was visited on a cold morning in week 60 of the campaign, and the filters were changed before the cold readings of that visit were written down.

Notes kept at the north gate on that cold day belong to section 11 of the logbook.

The river bridge sensor was visited on a mild morning in week 61 of the campaign, and the filters were changed before the mild readings of that visit were written down.

Notes kept at the river bridge on that mild day belong to section 11 of the logbook.

The bus depot sensor was visited on a dusty morning in week 62 of the campaign, and the filters were changed before the dusty readings of that visit were written down.

Notes kept at the bus depot on that dusty day belong to section 11 of the logbook.

The old market sensor was visited on a calm morning in week 63 of the campaign, and the filters were changed before the calm readings of that visit were written down.

Notes kept at the old market on that calm day belong to section 11 of the logbook.

The school yard sensor was visited on a windy morning in week 64 of the campaign, and the filters were changed before the windy readings of that visit were written down.

Notes kept at the school yard on that windy day belong to section 11 of the logbook.

The ring road sensor was visited on a foggy morning in week 65 of the campaign, and the filters were changed before the foggy readings of that visit were written down.

Notes kept at the ring road on that foggy day belong to section 11 of the logbook.

The rail yard sensor was visited on a humid morning in week 66 of the campaign, and the filters were changed before the humid readings of that visit were written down.

Notes kept at the rail yard on that humid day belong to section 12 of the logbook.

The east park sensor was visited on a dry morning in week 67 of the campaign, and the filters were changed before the dry readings of that visit were written down.

Notes kept at the east park on that dry day belong to section 12 of the logbook.

Chapter 5

Conclusions and Future Work

The monitoring system built in this thesis ran unattended for fourteen months and recorded air quality at nine sites with less than two percent of readings lost. Its measurements agreed with the reference station to within eight percent.

5.1 Limitations

The sensors were calibrated against a single reference station, so drift at the other sites could not be checked. The study covered one city and one winter.

5.2 Future Work

Future work should add a second reference station and run the network through a full year so that seasonal drift can be measured.

References

[1] Castell, N. et al. (2017). Can commercial low-cost sensor platforms contribute to air quality monitoring? *Environment International*, 99, 293-302.

[12] Author, A. and Author, B. (2012). A study of urban air quality sensors. *Journal of Environmental Monitoring*, 72, 100-110.

[13] Author, A. and Author, B. (2013). A study of urban air quality sensors. *Journal of Environmental Monitoring*, 73, 100-110.

[14] Author, A. and Author, B. (2014). A study of urban air quality sensors. *Journal of Environmental Monitoring*, 74, 100-110.

[15] Author, A. and Author, B. (2015). A study of urban air quality sensors. *Journal of Environmental Monitoring*, 75, 100-110.

[16] Author, A. and Author, B. (2016). A study of urban air quality sensors. *Journal of Environmental Monitoring*, 76, 100-110.

Appendix A

Calibration Tables

The harbour wall sensor was visited on a cold morning in week 78 of the campaign, and the filters were changed before the cold readings of that visit were written down.

Notes kept at the harbour wall on that cold day belong to appendix a of the logbook.

The town hall sensor was visited on a mild morning in week 79 of the campaign, and the filters were changed before the mild readings of that visit were written down.

Notes kept at the town hall on that mild day belong to appendix a of the logbook.

The north gate sensor was visited on a dusty morning in week 80 of the campaign, and the filters were changed before the dusty readings of that visit were written down.

Notes kept at the north gate on that dusty day belong to appendix a of the logbook.

The river bridge sensor was visited on a calm morning in week 81 of the campaign, and the filters were changed before the calm readings of that visit were written down.

Notes kept at the river bridge on that calm day belong to appendix a of the logbook.

The bus depot sensor was visited on a windy morning in week 82 of the campaign, and the filters were changed before the windy readings of that visit were written down.

Notes kept at the bus depot on that windy day belong to appendix a of the logbook.

The old market sensor was visited on a foggy morning in week 83 of the campaign, and the filters were changed before the foggy readings of that visit were written down.

Notes kept at the old market on that foggy day belong to appendix a of the logbook.

The school yard sensor was visited on a humid morning in week 84 of the campaign, and the filters were changed before the humid readings of that visit were written down.

Notes kept at the school yard on that humid day belong to appendix a of the logbook.

The ring road sensor was visited on a dry morning in week 85 of the campaign, and the filters were changed before the dry readings of that visit were written down.

Notes kept at the ring road on that dry day belong to appendix a of the logbook.

The rail yard sensor was visited on a rainy morning in week 86 of the campaign, and the filters were changed before the rainy readings of that visit were written down.

Notes kept at the rail yard on that rainy day belong to appendix a of the logbook.

The east park sensor was visited on a cold morning in week 87 of the campaign, and the filters were changed before the cold readings of that visit were written down.

Notes kept at the east park on that cold day belong to appendix a of the logbook.

The harbour wall sensor was visited on a mild morning in week 88 of the campaign, and the filters were changed before the mild readings of that visit were written down.

Notes kept at the harbour wall on that mild day belong to appendix a of the logbook.

The town hall sensor was visited on a dusty morning in week 89 of the campaign, and the filters were changed before the dusty readings of that visit were written down.

Notes kept at the town hall on that dusty day belong to appendix a of the logbook.

The north gate sensor was visited on a calm morning in week 90 of the campaign, and the filters were changed before the calm readings of that visit were written down.

Notes kept at the north gate on that calm day belong to appendix a of the logbook.